

MePo: Meta Post-Refinement for Rehearsal-Free General Continual Learning

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Core idea: meta training the pretrained model before deployment, then continually adapt with a stable reference geometry.

1. Introduction

General continual learning (GCL): real-world scenarios continual learning with **online data streams** and **blurry task boundaries**, making it increasingly difficult for AI models to rapidly capture and effectively balance successive information^[1,2].

Pretrained models (PTMs) in CL: parameter-efficient tuning (PET) for representation learning, and recover old task distributions in the representation space for output alignment^[3].

Biologic inspiration: (1) **meta-plasticity**^[4]: biological brain retain substantial “pre-trained knowledge”, positioning them in a critical state of neuro dynamics for rapid adaptation; (2) **reconstructive memory**^[5]: the neural representations of incoming memories are continually encoded into and reconstructed from a shared representation space that enables real-time generalization.

2. Contribution

(1) Meta Rep. learns a GCL-ready initialization.

(2) Meta Cov. stabilizes feature geometry online.

Strength: Plug-and-play, rehearsal-free, reusable, generalizable, and no extra computational overhead.

7. Computational overheads

Phase 1: meta post-training

Training	Image Size	Epochs	Batch Size	Total Steps	Images Processed
Sup-21K (Pretraining) (Dosovitskiy et al., 2020a)	224×224	90	4096	~310k	~1.3B
Sup-21K (MePo Post-Refinement)	224×224	50	256	~12.8k	~2M

Meta epoch	1	2	3	4	5	Average
Time (mins)	14.43	14.20	14.13	14.00	13.96	14.14

Phase 2: general continual learning

Method	+Param.	+Ratio	Time	Accuracy
MVP [19]	639k	0.74%	5.34s	41.50
w/ MePo	1215k	1.41%	5.34s	46.35
MISA [12]	637k	0.74%	4.84s	51.52
w/ MePo	1213k	1.41%	4.84s	54.86

8. Reference

- [1] Pietro Buzzega, Matteo Boschini, Angelo Porrello, Davide Abati, and Simone Calderara. Dark experience for general continual learning: a strong, simple baseline. *NeurIPS*, 33:15920–15930, 2020.
- [2] Jun-Young Moon, Keon-Hee Park, Jung Uk Kim, and Gyeong-Moon Park. Online class incremental learning on stochastic blurry task boundary via mask and visual prompt tuning. In *ICCV*, pages 11731–11741, 2023.
- [3] Da-Wei Zhou and Hai-Long Sun and Jingyi Ning and Han-Jia Ye and De-Chuan Zhan. Continual Learning with Pre-Trained Models: A Survey. *IJCAI* 2024.
- [4] Wickliffe C Abraham. Metaplasticity: tuning synapses and networks for plasticity. *Nature342Reviews Neuroscience*, 9(5):387–387, 2008.343.
- [5] Bo Lei, Bilin Kang, Yuejun Hao, Haoyu Yang, Zihan Zhong, Zihan Zhai, and Yi Zhong. Reconstructing a new hippocampal engram for systems reconsolidation and remote memory381updating. *Neuron*, 2024.

3. Method Overview

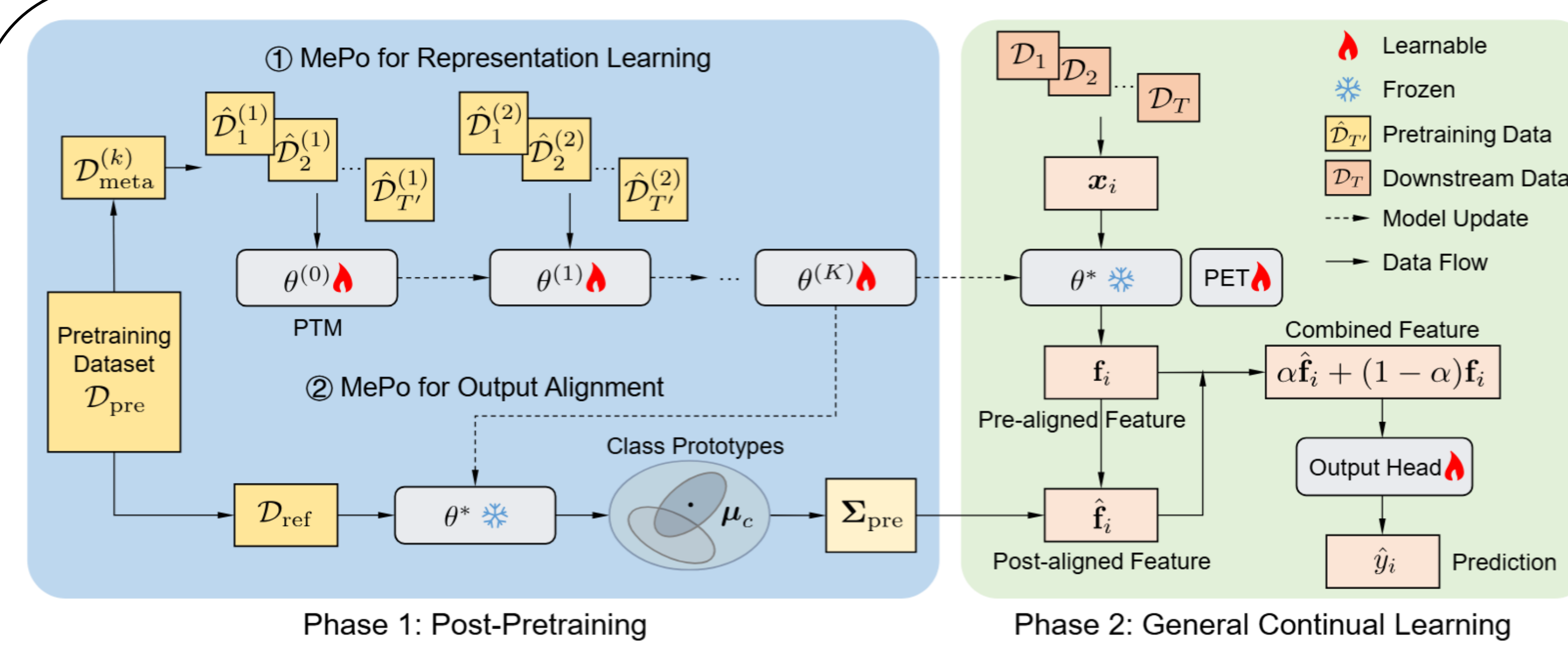


Fig. 1 The proposed MePo framework.

4. Main Results

PTM	Method	CIFAR-100		ImageNet-R		CUB-200	
		$A_{AUC}(\uparrow)$	$A_{Last}(\uparrow)$	$A_{AUC}(\uparrow)$	$A_{Last}(\uparrow)$	$A_{AUC}(\uparrow)$	$A_{Last}(\uparrow)$
Sup-21K	Seq FT	19.71±3.39	10.42±4.92	7.51±3.94	2.29±0.85	3.47±0.41	1.49±0.42
	Linear Probe	49.69±6.09	23.07±7.33	29.24±1.26	16.87±3.14	28.96±2.46	17.33±3.08
	Seq FT (SL) [38]	64.90±7.18	62.06±1.89	47.20±1.47	39.60±2.43	56.16±4.32	56.50±3.08
	CODA-P [25]	78.81±3.38	80.30±1.58	50.11±2.14	46.17±2.00	64.96±3.30	59.28±3.14
	Deep L2P [33]	78.12±0.61	77.73±1.09	42.39±0.23	38.16±1.37	60.95±1.22	56.31±2.53
	w/ MePo (Ours)	83.63 ±0.61	83.98 ±0.29	58.71 ±1.28	55.13 ±1.16	64.92 ±1.47	63.30 ±1.52
	DualPrompt [32]	66.36±4.42	58.09±4.40	38.63±2.19	30.71±0.82	55.73±2.77	47.08±4.94
	w/ MePo (Ours)	71.37±4.07	66.48±2.82	44.65±2.09	36.76±1.21	58.36±2.59	52.16±3.74
	MVP [19]	68.13±4.34	60.56±2.57	41.50±1.15	34.14±3.95	56.78±2.88	50.25±3.53
	w/ MePo (Ours)	72.18±4.50	68.45±1.59	46.35±1.31	38.21±3.66	58.73±3.31	52.22±2.80
Sup-21/1K	MISA [12]	80.35±2.39	80.75±1.24	51.52±2.09	45.08±1.43	65.40±3.01	60.20±1.82
	w/ MePo (Ours)	82.30 ±2.83	83.99 ±1.35	54.86 ±2.20	49.18 ±1.38	68.13 ±3.17	64.75 ±1.00
	Deep L2P [33]	69.15±1.66	68.57±1.38	42.74±0.83	39.22±2.14	39.20±1.69	46.76±1.87
	w/ MePo (Ours)	78.75 ±1.18	77.52 ±1.03	62.71 ±0.99	58.91 ±0.08	48.36 ±1.88	50.88 ±2.85
	DualPrompt [32]	64.84±2.62	67.22±8.54	49.52±2.92	47.14±3.39	43.96±2.00	41.20±7.61
	w/ MePo (Ours)	67.18±4.48	57.95±3.69	54.75±1.66	44.75±0.74	47.06±3.19	38.24±9.29
	MVP [19]	65.26±3.87	53.66±5.61	51.26±1.47	41.41±4.81	45.12±3.08	37.95±9.32
	w/ MePo (Ours)	70.25±4.23	62.05±2.39	61.28±1.21	50.82±3.70	49.72±3.53	42.81±6.74
	MISA [12]	62.91±7.96	67.99±7.41	50.87±1.69	47.75±2.87	42.76±2.33	44.05±1.94
	w/ MePo (Ours)	78.01±3.09	76.73±1.06	64.23 ±1.30	58.20 ±0.51	55.31 ±4.52	56.58 ±2.33
iBOT-21K	Deep L2P [33]	64.48±1.23	66.71±1.27	33.68±2.78	36.24±1.83	16.22±0.85	27.14±0.75
	w/ MePo (Ours)	75.83 ±1.23	76.40 ±0.94	55.30 ±0.50	52.38 ±1.87	40.90 ±1.44	46.50 ±2.90
	DualPrompt [32]	63.09±2.36	61.20±8.76	41.33±2.11	35.58±3.24	24.56±2.25	21.32±6.38
	w/ MePo (Ours)	65.76±3.56	59.21±3.18	48.06±2.20	37.69±2.10	38.19±3.74	31.03±11.55
	MVP [19]	64.01±3.27	50.00±11.45	43.89±1.88	34.19±4.56	29.59±3.28	27.85±8.89
	w/ MePo (Ours)	66.88±4.86	57.19±2.63	53.75±1.38	42.55±3.08	40.99±3.45	34.66±8.40
	MISA [12]	65.30±2.28	67.43±6.75	40.94±1.22	36.16±1.58	18.62±3.36	23.66±2.21
	w/ MePo (Ours)	75.80±3.77	76.02±1.18	57.00 ±2.52	49.86 ±1.22	49.33 ±3.59	45.68 ±2.59

5. Ablation

PTM	Meta Rep	Meta Cov	ImageNet-R (MVP)		ImageNet-R (MISA)		CUB-200 (MVP)		CUB-200 (MISA)	
			$A_{AUC}(\uparrow)$	$A_{Last}(\uparrow)$	$A_{AUC}(\uparrow)$	$A_{Last}(\uparrow)$	$A_{AUC}(\uparrow)$	$A_{Last}(\uparrow)$	$A_{AUC}(\uparrow)$	$A_{Last}(\uparrow)$
Sup-21K	✓	✗	41.50±1.15	34.14±3.95	51.52±2.09	45.08±1.43	56.78±2.88	50.25±3.53	65.40±3.01	60.20±1.82
	✓	✗	46.32±1.29	38.06±3.77	52.35±2.09	45.81±1.08	58.67±2.80	51.65±3.23	64.83±2.82	59.57±1.73
	✗	✓	40.51±1.20	33.99±4.11	53.59±2.26	47.90±1.65	55.82±3.64	51.49±2.72	68.03±3.05	65.30 ±1.82
	✓	✓	46.35 ±1.31	38.21 ±3.66	54.86 ±2.20	49.18 ±1.38	58.73 ±3.31	52.22 ±2.80	68.13 ±3.17	64.75±1.00
Sup-21/1K	✗	✗	51.26±1.47	41.41±4.81	50.87±1.69	47.75±2.87	45.12±3.08	37.95±9.32	42.76±2.33	44.05±1.94
	✓	✗	57.50±1.18	46.75±4.85	56.71±1.08	50.29±2.04	47.26±3.38	39.65±8.09	44.68±2.46	43.88±2.72
	✗	✓	55.83±1.62	45.83±5.07	57.66±0.96	52.30±0.58	47.69±3.05	40.51±8.67	49.60±3.31	47.21±1.72
	✓	✓	61.28 ±1.21	50.82 ±3.70	64.23 ±1.30	58.20 ±0.51	49.72 ±3.53	42.81 ±6.74	55.31 ±4.52	56.58 ±2.33
iBOT-21K	✗	✗	43.89±1.88	34.19±4.56	40.94±1.22	36.16±1.58	29.59±3.28	27.85±8.89	18.62±3.36	23.66±2.21
	✓	✗	52.67±1.41	41.91±3.95	50.21±1.93	43.52±1.39	40.61±3.41	34.17±9.09	39.44±2.93	40.38±1.79
	✗	✓	47.44±1.76	37.02±5.00	44.24±1.90	38.08±1.15	31.75±3.43	30.35±9.12	20.65±3.22	22.92±0.75
	✓	✓	53.75 ±1.38	42.55 ±3.08	57.00 ±2.52	49.86 ±1.22	40.99 ±3.45	34.66 ±8.40	49.33 ±3.59	45.68 ±2.59

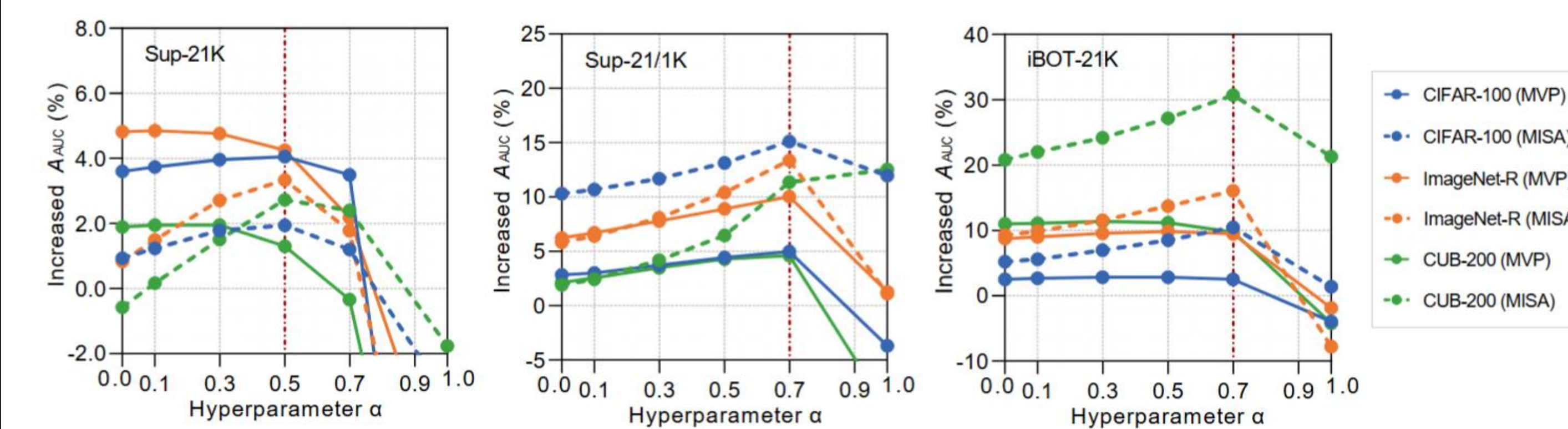


Fig. 3 Empirical evaluation of the combination weight α in MePo.

6. Analysis: MePo obtains more separable and sparse representations.

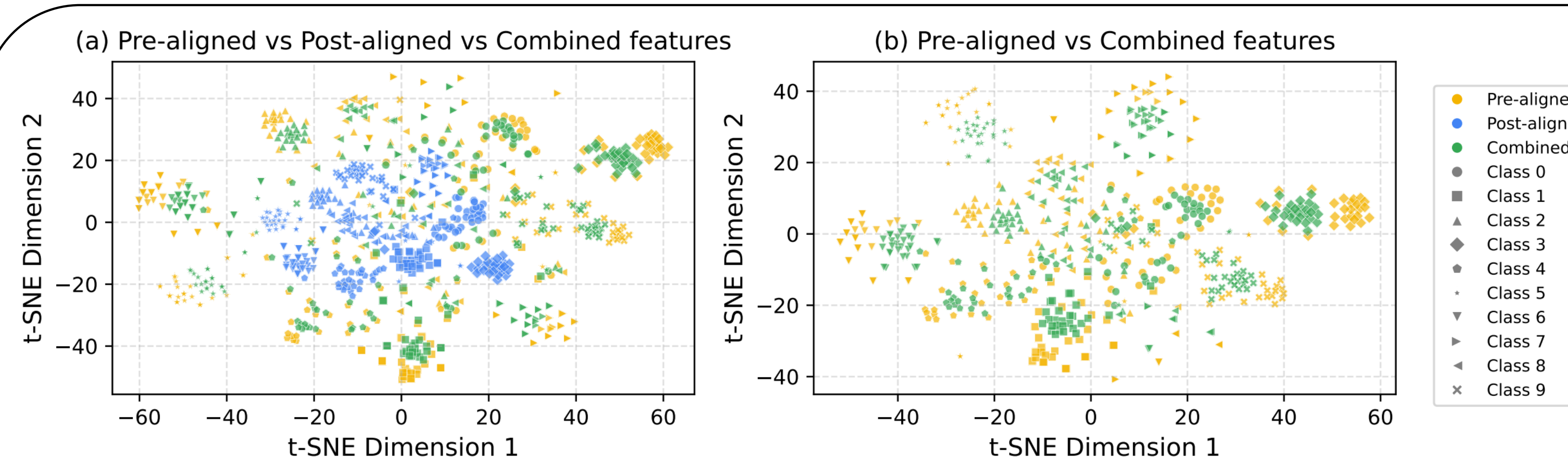


Fig. 4 Visualization of pre-aligned, post-aligned, and combined features with t-SNE. Here we take the setup of MISA w/ MePo, ImageNet-R, and Sup-21/1K as an example.

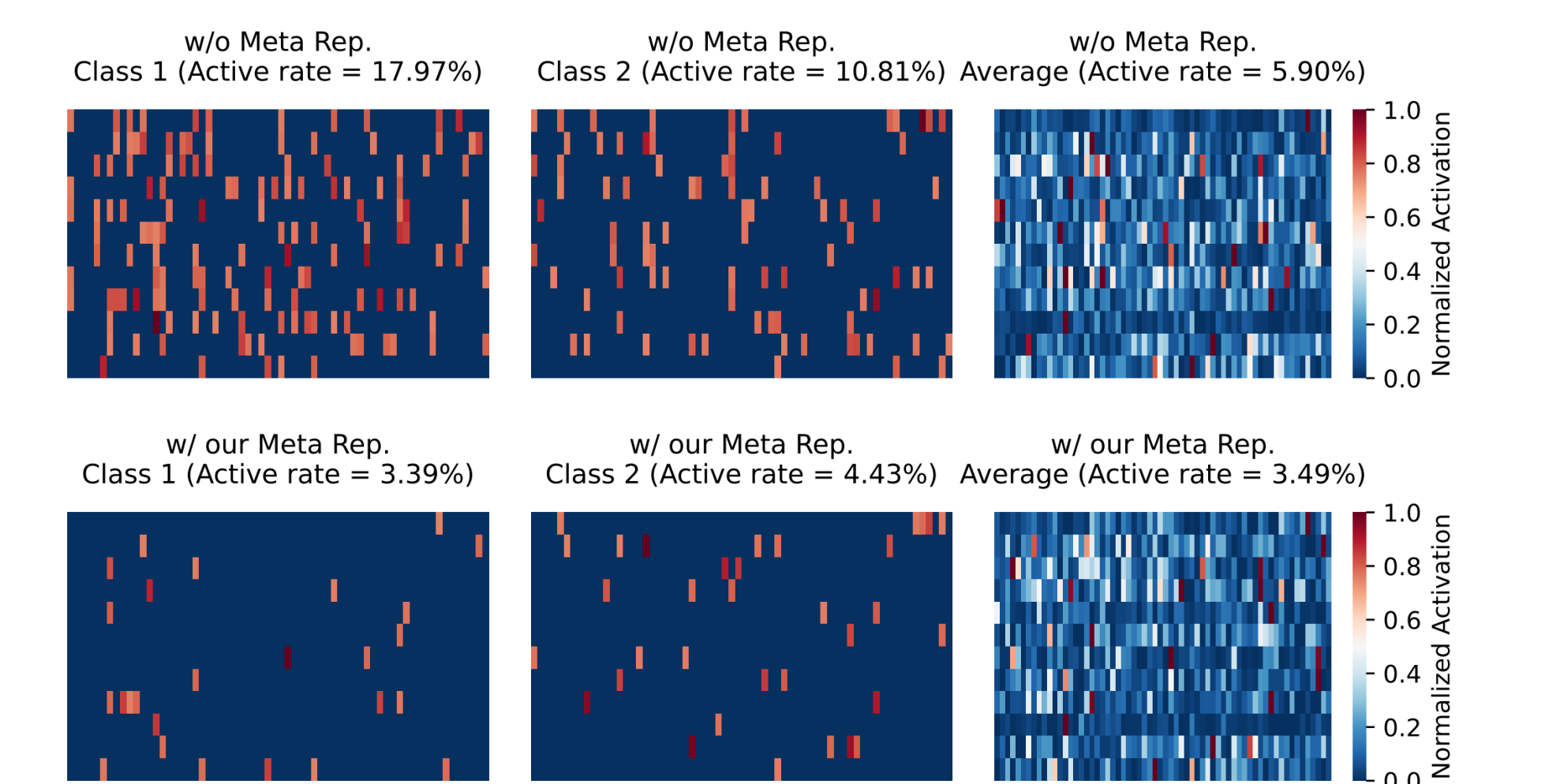


Fig. 5 Visualization of class-wise prototypes on ImageNet-R under Sup-21K.